

Assignment 3

**Day 3: BIOS and PIST Troubleshooting Theory(30 min):
Understanding BIOS and POST errors. Practical(30 min):
Analyze POST error codes on a given setup.**

Theory: Understanding BIOS and POST Errors

BIOS (Basic Input/Output System) is firmware stored on the motherboard. It starts and checks the basic hardware when the computer is switched on.

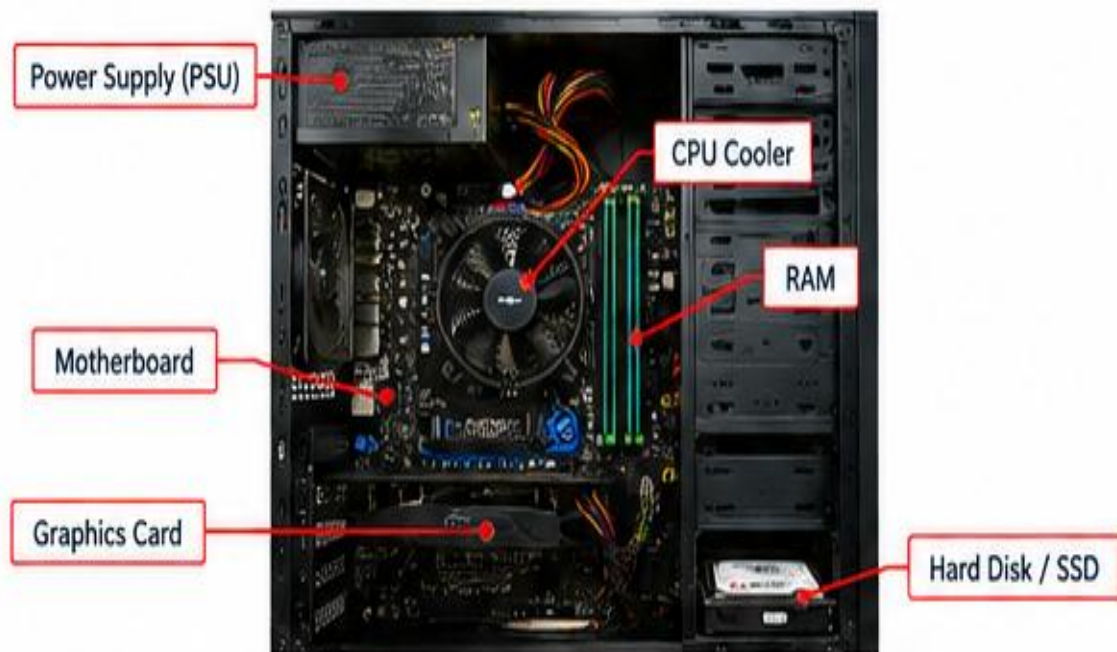
POST (Power-On Self-Test) is a process performed during startup to check important hardware such as RAM, CPU, keyboard and display. If a hardware problem is detected, BIOS may show an error message or produce beep codes.

Task 1: Common BIOS or POST Error Messages

The following are five common BIOS or POST error messages:

No	BIOS/POST Error Message	Usually Indicates
1	CMOS checksum error – Defaults loaded	CMOS/BIOS settings may be corrupted or the CMOS battery may be weak.
2	Keyboard not detected / Keyboard error	Keyboard is disconnected, faulty, or there may be a problem with the keyboard port.
3	CPU Fan Error	CPU fan may not be connected, may be blocked, or may not be spinning properly.
4	No boot device found	SSD/HDD may not be detected, the boot order may be incorrect, or no bootable operating system is available.

5	Memory error / Memory test failure	RAM may be loose, incompatible, or faulty.
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Task 2: Three Beeps on AMI BIOS

Given Scenario

A PC gives **3 beeps in a repeating pattern** during startup and there is no display on the monitor.

For **classic AMI BIOS beep-code documentation**, **3 short beeps** typically indicate a **base 64 KB memory failure**, which points toward a problem with the system RAM/memory subsystem.

Troubleshooting Step 1 – Reseat the RAM

1. Turn off the PC and disconnect the power.
2. Open the computer case.
3. Remove the RAM module carefully.
4. Reinsert the RAM firmly into its slot.
5. Start the computer and check whether the display returns.

Troubleshooting Step 2 – Test the RAM

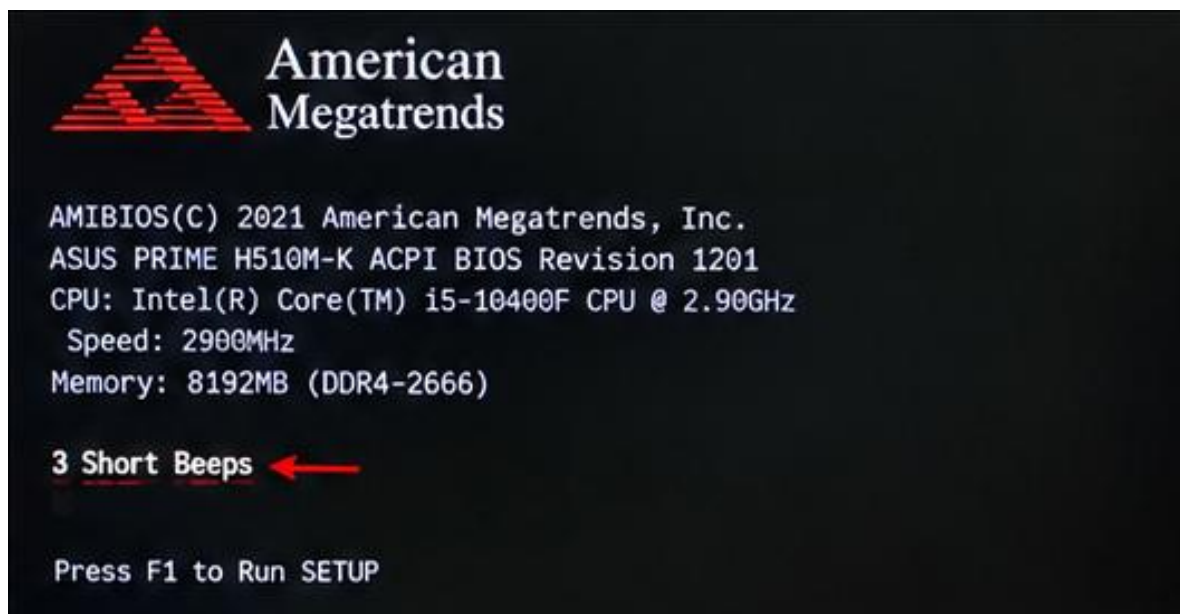
If the problem continues:

1. If there are multiple RAM sticks, test them **one at a time**.
2. Try another compatible RAM slot if appropriate.
3. Check the RAM for physical damage.
4. If one module consistently causes the problem, it may need replacement.

Conclusion

Three short AMI BIOS beeps can indicate a **memory/RAM-related failure**. Reseating and testing the RAM are two basic troubleshooting steps.

Note: Beep codes can vary between BIOS versions and motherboard manufacturers, so the motherboard's manual should be checked for the exact code.



Task 3: Access BIOS Setup Utility

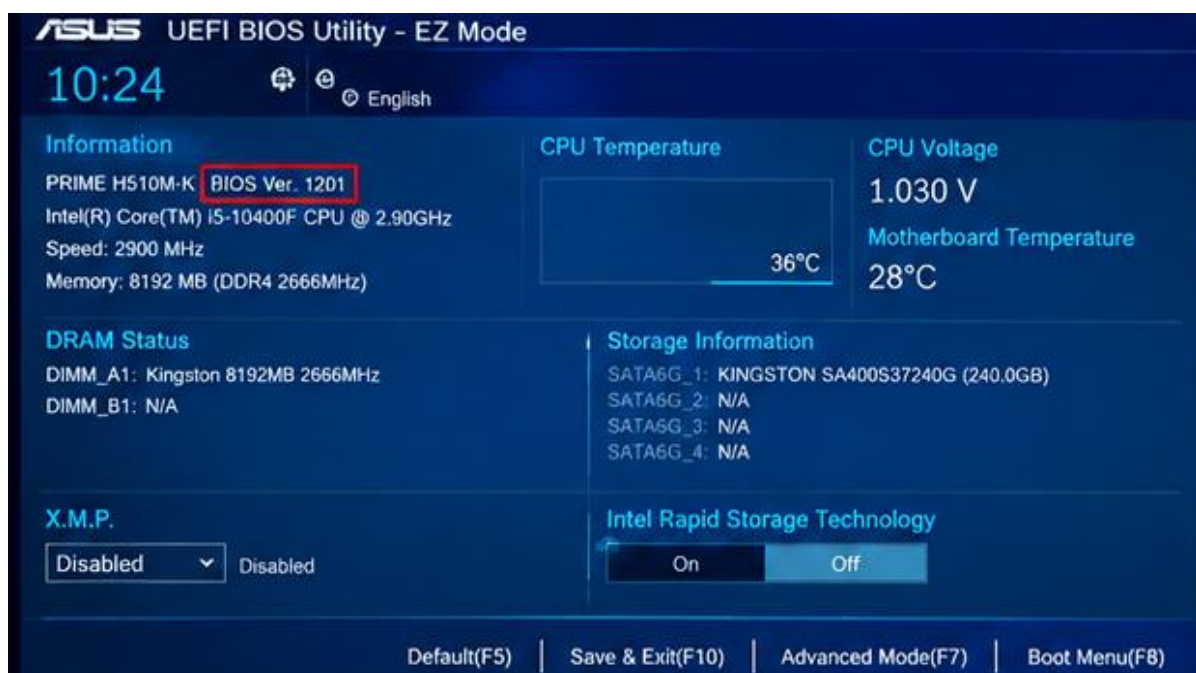
To access the BIOS setup utility and identify the motherboard manufacturer and BIOS version.

Procedure

1. Start or restart the computer.
2. During startup, repeatedly press the BIOS setup key.
3. Common keys are **Delete**, **F2**, **F10** or **Esc**, depending on the computer.
4. The BIOS/UEFI setup screen will appear.
5. Open the **Main / Information** section if necessary.
6. Note the **motherboard manufacturer** and **BIOS version**.
7. Take a screenshot of the BIOS main screen.

Motherboard Manufacturer: ASUS (replace with your actual motherboard manufacturer)

BIOS Version: 1201 (replace with your actual BIOS version)



Task 4: CMOS Checksum Error – Defaults Loaded

Error Message

“CMOS checksum error – Defaults loaded.”

This message means that the BIOS has detected that the **CMOS settings are invalid, corrupted, or different from the previously saved settings.**

The CMOS memory stores BIOS configuration information such as:

- Date and time
- Boot order
- Hardware configuration
- Other BIOS settings

A **weak or dead CMOS battery** is one common reason for this error.

How to Resolve the Problem

Step 1 – Enter BIOS Setup :Restart the computer and enter BIOS using the appropriate key, such as **Delete or F2.**

Step 2 – Load default settings: Select **Load Optimized Defaults / Load Setup Defaults** and save the settings.

Step 3 – Check date and time: Set the correct date and time if they have been reset.

Step 4 – Check the CMOS battery: If the error returns repeatedly, switch off the computer and disconnect power. On a typical desktop motherboard, the CMOS battery is a **CR2032 coin-cell battery.**

If the battery is weak/dead, it can be replaced with a compatible battery.

Step 5 – Save and restart: Save the BIOS settings and restart the computer.

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AMIBIOS(C)2001 American Megatrends, Inc.
BIOS Date: 02/22/06 20:54:49 Ver: 08.00.02

Press DEL to run Setup
Checking NVRAM..

512MB OK
Auto-Detecting Pri Channel (0)...IDE Hard Disk
Auto-Detecting Pri Channel (1)...Not Detected
Auto-Detecting Sec Channel (0)...CDROM
Auto-Detecting Sec Channel (1)...Not Detected

CMOS Checksum Bad
CMOS Date/Time Not Set
Press F1 to Run SETUP
Press F2 to load default values and continue

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